

20. QUESTION D1: MESODERM IMPLANTATION

DURATION : 02:33

STUDY SHEET

COURSE SUMMARY

In this session, you will learn the crucial steps of mesoderm formation within the embryo, a process orchestrated by wave-like movements. The focus is on the dynamics of the primitive streak and Hensen's node, which are fundamental for cellular development. You will discover how the notochord forms in synchrony with the regression of the primitive streak, and how the epithelial cells of the epiblast contribute to the emergence of the intermediate tissue, the mesoblast, which will become the mesoderm. This understanding will enrich your practice in embryology and osteopathy, offering valuable tools for comprehending embryonic development.

MESODERM FORMATION

The **mesoderm** forms through a **wave** movement that takes shape within the embryo. This movement creates a **field of aspiration** at the primitive streak, which is a key element in embryonic development.

As the **primitive streak** regresses, it descends with **Hensen's node** until it completely disappears. This dynamic is represented by a superior view, where the **amniotic cavity** can be observed above.

The **notochord** forms in response to this movement. At this stage, a traction occurs at the **pedicle**, leading to the appearance of the primitive streak. It is important to note that the formation of the notochord and the primitive streak occurs **simultaneously**, although there is a fine chronology in this process.

The illustrated drawing shows a section highlighting the movement of the **epithelial cells** of the epiblast. These cells are drawn towards the centre, thus filling the space created between the epiblast and the hypoblast.

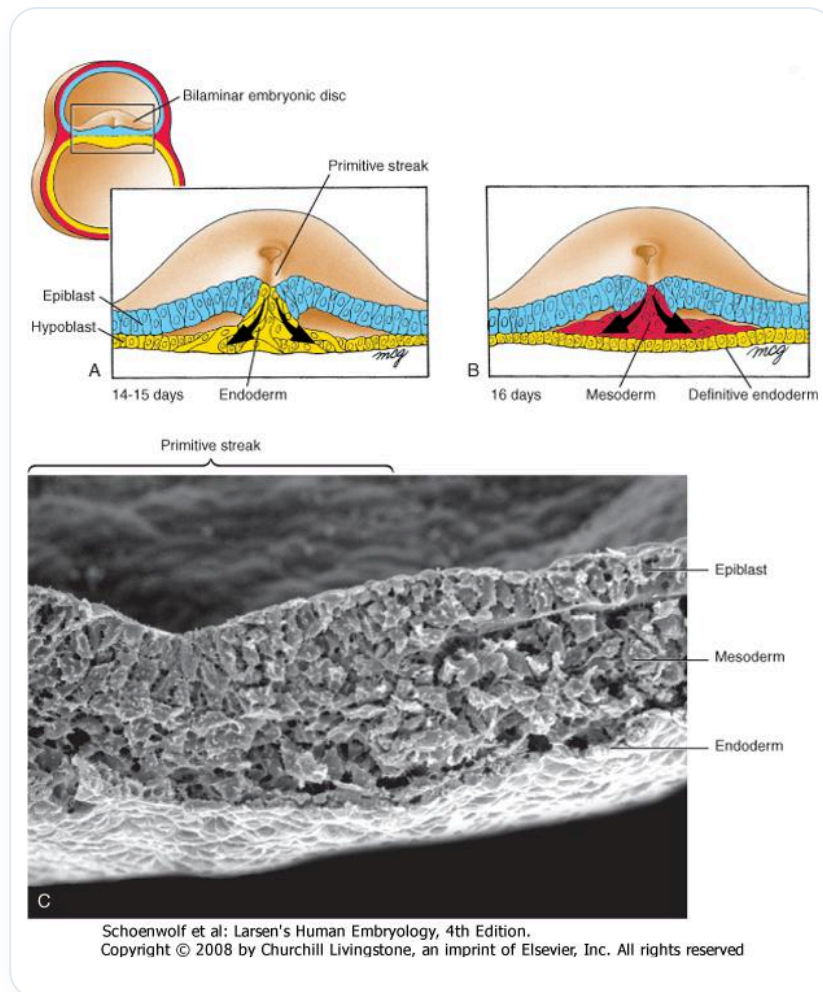


FIG. — Gastrulation

This space, which opens during the formation of the wave, is essential for the appearance of the **intermediate tissue** known as the **mesoblast**, which will become the future **mesoderm**.

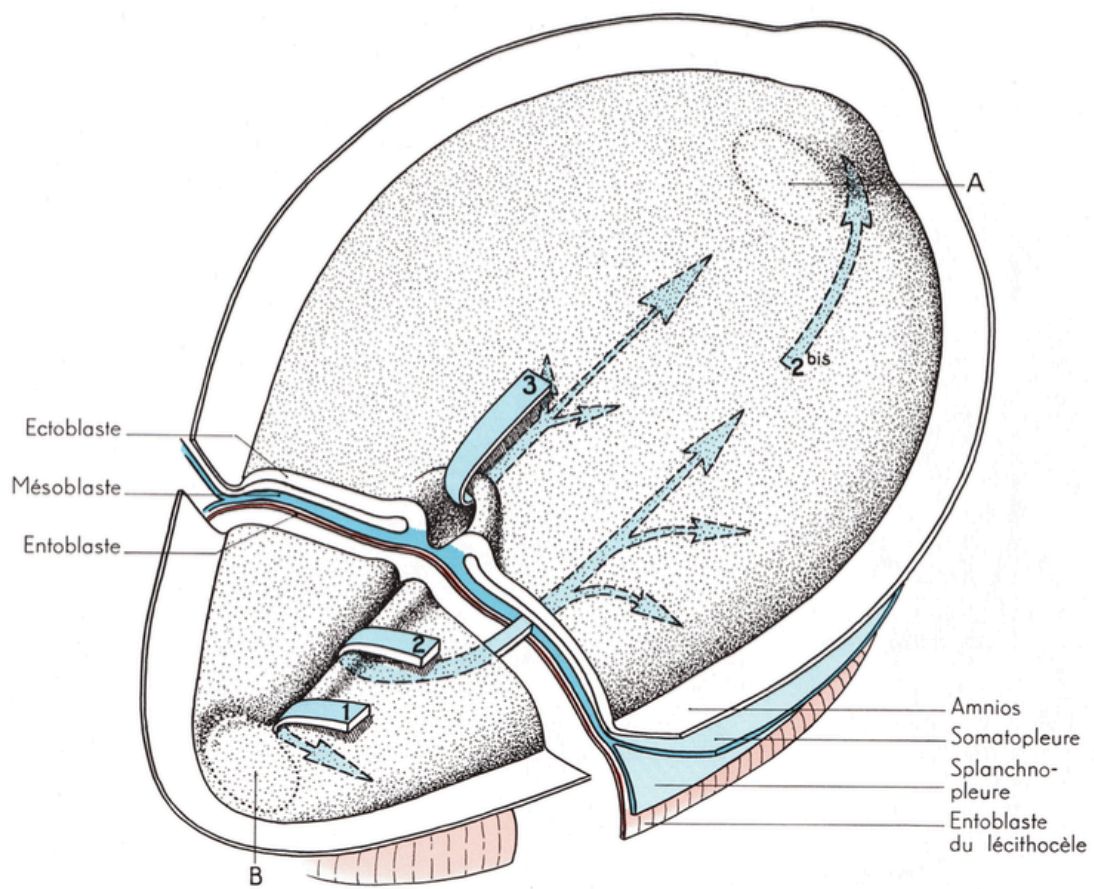


FIG. — Avian gastrulation

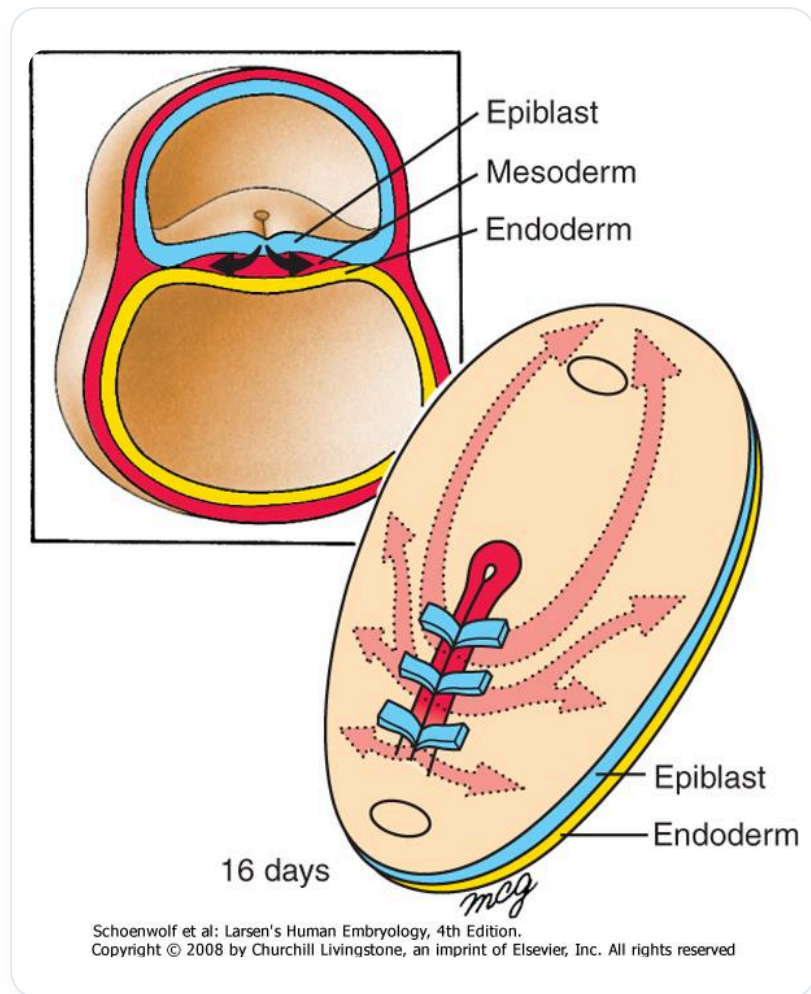


FIG. — Human gastrulation 16 days